

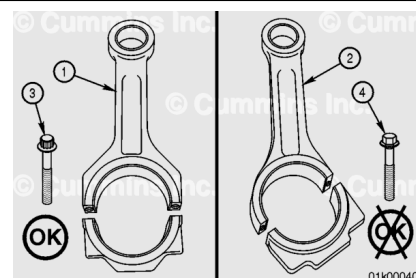
Trainings ▾

Install

The following steps **must** be used for engines containing the straight split connecting rod (1)

Use the following procedure for engines containing the angle split rod (2). Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html)

Note : The straight split connecting rod (1) can be identified by 12 point head capscrews (3).



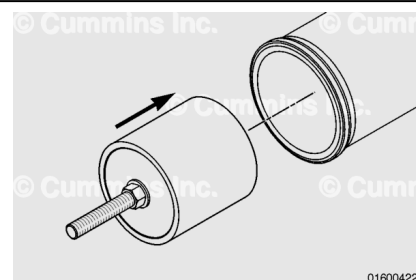
⚠ CAUTION ⚠

Do not allow the connecting rod to impact the liner during the following steps, or liner cracking or other damage may occur.

Remove any excess oil from the liner bore.

Insert the plunger, power cylinder install/removal tool, Part Number 4919227, into the liner.

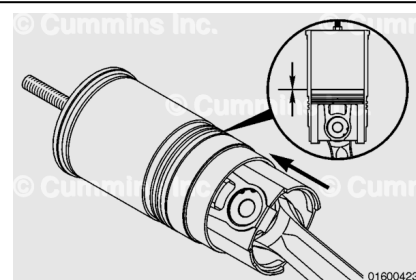
The top face of the tool should be approximately 20 mm [0.8 in] below the top of the liner.



Push the piston and connecting rod into the cylinder liner until the piston contacts the bottom of the plunger.

Tighten the plunger to lock the plunger into the liner.

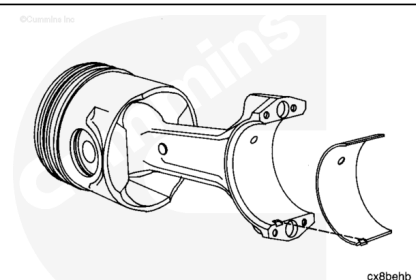
Torque Value: 125 n·m [92 ft-lb]



Use a lint-free cloth to clean the connecting rod and bearing shells.

Install the connecting rod bearing.

Position the tang as shown in the illustration. The end of the bearing **must** be even with the cap mounting surface.

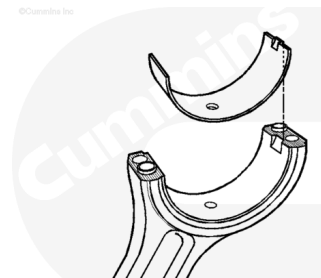


The bearings **must** be installed in their original location if new bearings are **not** used. All of the rod bearings are identical.

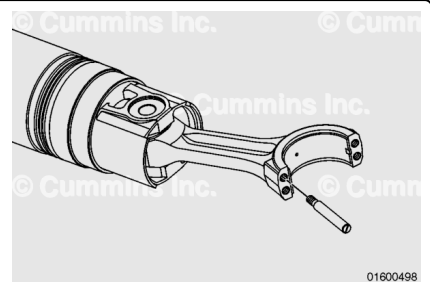
⚠ CAUTION ⚠

To reduce the possibility of engine damage, do not lubricate the back side of the bearing shells.

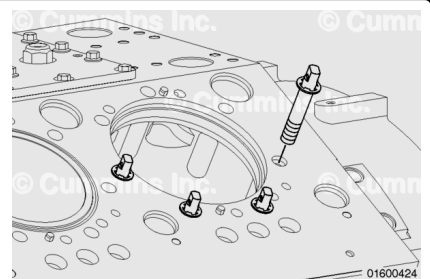
Lubricate the bearing surfaces with Lubriplate™ 105, Part Number 3163086.



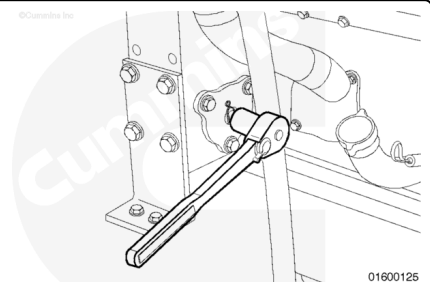
Install the four connecting rod guide studs, Part Number 4918922, into the connecting rod.



Insert the four block guide studs into the block at the locations shown until the circlip contacts the block face.

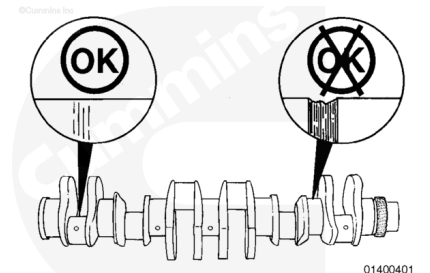


Rotate the crankshaft until the journal for the connecting rod being installed is at top dead center and centered in the cylinder bore.



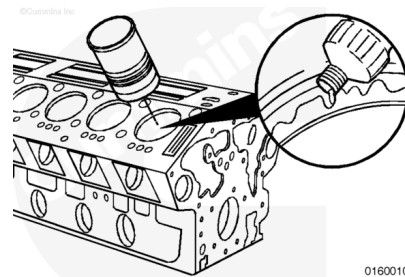
Note : The crankshaft is shown out of the engine for clarity.

Check the crankshaft rod journals for damage.



Do **not** use petroleum-based oil, as this will cause swelling of the packing ring.

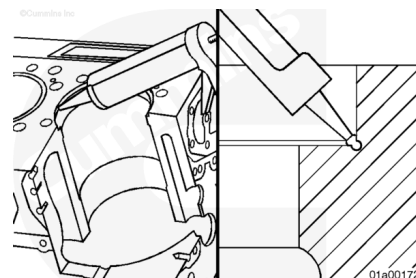
Use vegetable oil to lubricate the inside diameter of the packing ring bores.



Apply a 2 mm [0.079 in] bead of gasket sealant, Part Number 4918771, to the corner of the cylinder block counterbore.

The end of the bead **must** overlap the beginning of the bead by approximately 6 mm [0.236 in].

Install the cylinder liner within 15 minutes of applying the sealant.



⚠ WARNING ⚠

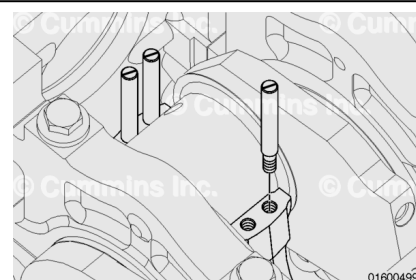
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



The connecting rod **must not** contact the block and **must** be correctly aligned with the crankshaft journal during installation.

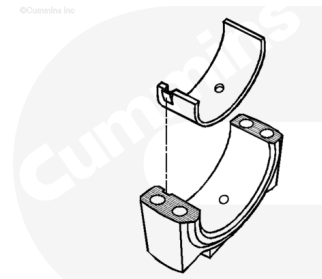
With assistance, or with a hoist, slowly lower the power cylinder into the cylinder bore until the connecting rod contacts the crankshaft journal.

Remove the connecting rod guide studs.



⚠ CAUTION ⚠

The connecting rods and caps are not interchangeable. The rods and the caps are machined as an assembly. Failure will result if they are mixed.



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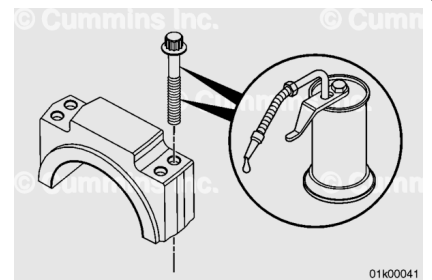
Install the lower bearing shell in the rod cap. Make sure the tang of the bearing shell is in the slot of the cap and the end of the bearing is even with the cap surface.

Lubricate the bearing surfaces with Lubriplate™ 105, Part Number 3163086

Lubricate the capscrew head flange with clean gear oil (SAE W140).

Lubricate the capscrew threads with clean engine oil (SAE 15W-40).

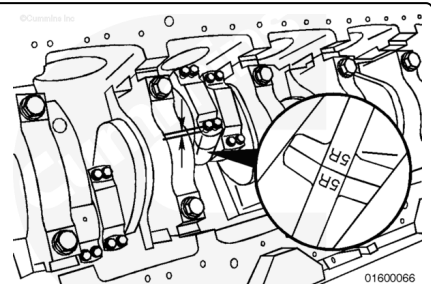
Install the capscrews into the caps.



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⚠ CAUTION ⚠

The connecting rod and connecting rod cap are a matched set. The cylinder number on the rod and cap must be the same. Damage to the engine can occur if the rods and rod caps are mixed.



01600066

⚠ CAUTION ⚠

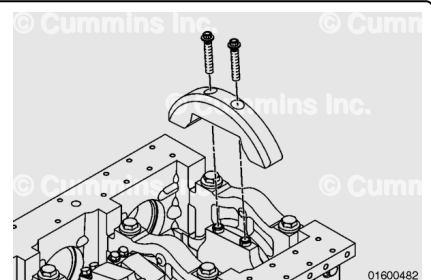
The side of the connecting rod with the bearing tang must be on the inboard side of the engine. If the connecting rods are installed backwards, engine damage will occur.

Install the connecting rod cap. Tighten the capscrews alternately and evenly to pull the cap over the dowel pins.

Do **not** tighten the capscrews at this stage.

If a counterweight was previously removed, follow the instructions in the Clean and Inspect for Reuse section. The counterweight should be installed at this time. Refer to Procedure 001-016 in Section 1.

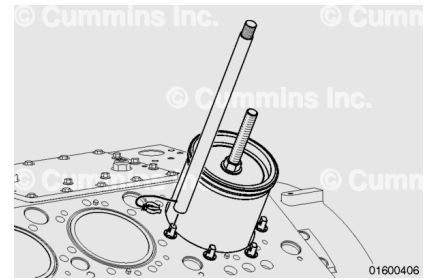
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Attach the four frame tube legs to the lower guide studs and secure them with locking pins.

The fixture used in the following steps is used to aid with the alignment of the liner into the block.

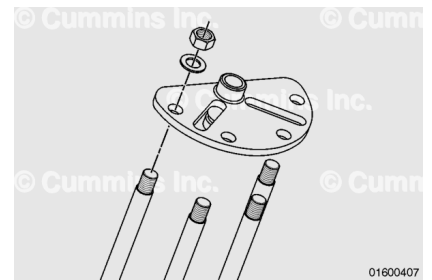


⚠ CAUTION ⚠

Do not overtighten the nuts or tool damage may occur.

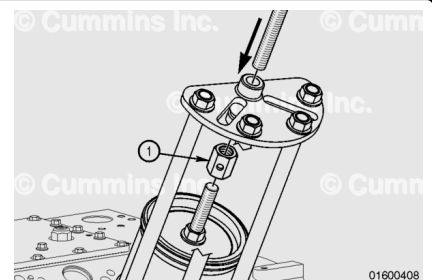
Attach the top plate with the boss facing upward.

Secure the plate with the four nuts and washers provided.



During the following tool assembly, and until instructed otherwise, the upper nut and thrust washer **must not** be fitted.

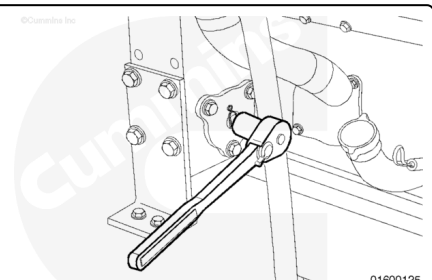
Insert the threaded bar through the top plate and attach it to the plunger using the hexagonal coupling (1).



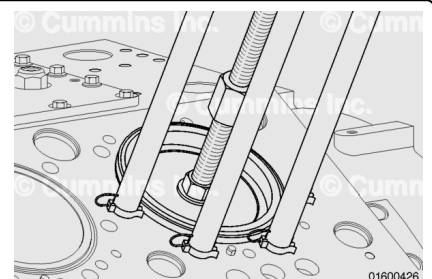
During the following procedure, the cylinder liner should be drawn into the block. If no movement is observed, stop and investigate what is preventing the liner from being drawn into the block.

Do **not** attempt to draw the liner flush with the block.

Use an engine barring mechanism to rotate the engine to lower the power cylinder assembly into the block.



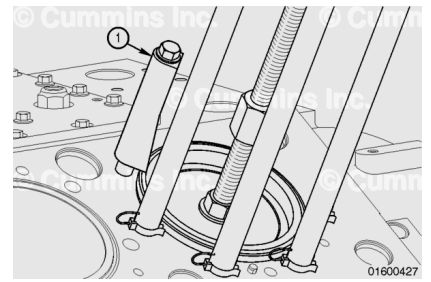
Lower the power cylinder until the liner contacts the press fit area of the block, as illustrated.



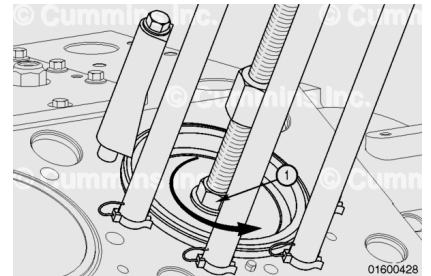
Do **not** attempt to press the liner into the press fit area using liner clamps.

To prevent liner movement during plunger extraction, use a liner clamp set (1), Part Number 3822503, and two cylinder head capscrews to hold the liner in position, as illustrated.

Hand-tighten the capscrews.

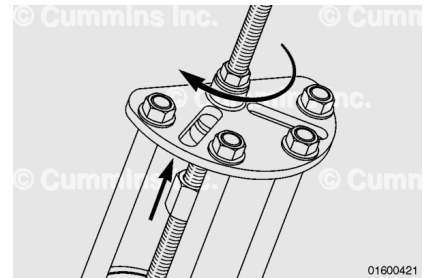


Loosen the plunger nut (1).



Fit the thrust washer and nut to the threaded rod.

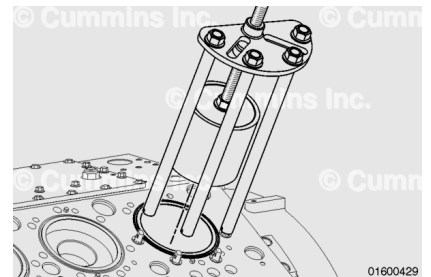
Continue rotating the nut **clockwise** to draw the plunger from the cylinder liner to a position where it can be easily removed by hand.



Release the locking pins from the block guide studs and remove the power cylinder tool as a complete assembly.

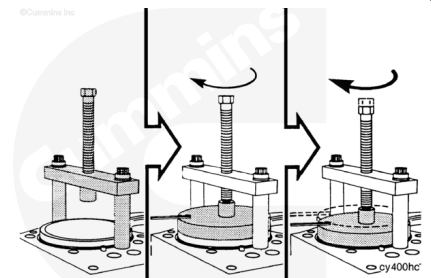
Remove the block guide studs.

Remove the liner clamps.



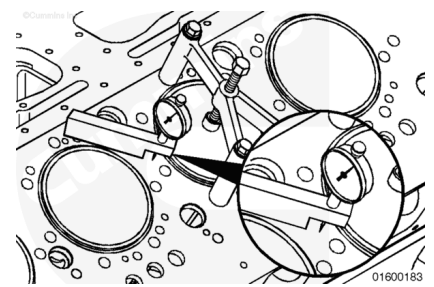
Use a liner installation tool, Part Number 3163329, or equivalent, to fully install the cylinder liner. Refer to Procedure 001-028 in Section 1.

(</qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html>)



Measure the cylinder liner protrusion.

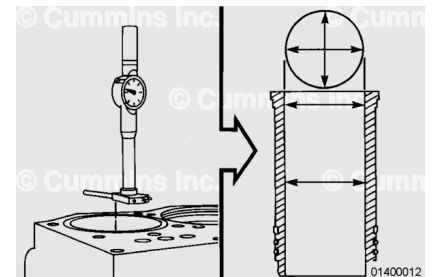
Cylinder Liner Protrusion		
mm		in
0.15	MIN	0.006
0.22	MAX	0.009



If the protrusion is **not** correct, the liner **must** be removed. The protrusion can be adjusted using seal rings and/or machining the counterbore ledge. Contact a Cummins® Authorized Repair Location if machining is required.

Use a dial bore gauge to measure the inside diameter of the cylinder liner at the top and middle of the cylinder liner.

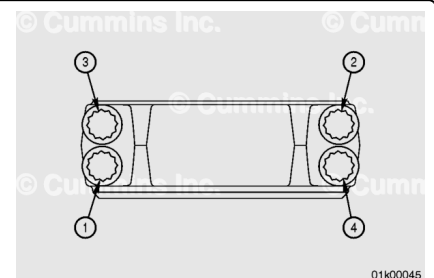
Perform two measurements at each location. The measurements **must** be 90-degrees apart.



New Cylinder Liner Inner Diameter		
mm		in
158.75	MIN	6.250
158.78	MAX	6.251

The inside diameter **must not** be more than 0.076 mm [0.003 in] out of round.

Note : Use the following steps for engines containing the straight split connecting rod **only**. Use the following procedure for engines containing the angle split connecting rod. Refer to Procedure 001-054 in Section 1 (/qs3/pubsys2/xml/en/procedures/56/56-001-054-tr.html). The straight split connecting rod can be identified by 12 point head capscrews.



Tighten the four connecting rod bearing cap capscrews.

Use sequence 1,2,3,4 on each connecting rod when tightening the connecting rod capscrews. See the illustration.

Torque Value:

1. 115 n•m [85 ft-lb]
2. Repeat step 1
3. Loosen all capscrews at least two revolutions
4. 95 n•m [70 ft-lb]
5. Repeat step 4
6. Advance 60 degrees

Use the following tightening procedure if accessing the connecting rod cap through the hand hole cover and a torque angle dial is inaccessible.

Prior to applying this process, be sure that all capscrews have been tightened by hand as far as possible to prevent thread damage or mis-engagement.

Use the following steps and sequence to tighten the capscrews.

Use sequence 1, 2, 3, 4 on each connecting rod when tightening the connecting rod capscrews. See the illustration.

Torque Value:

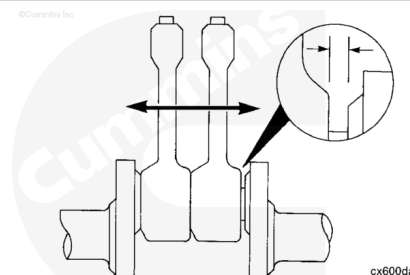
1. 115 n•m [85 ft-lb]
2. Repeat Step 1
3. Loosen all capscrews at least two revolutions
4. 95 n•m [70 ft-lb]
5. Repeat Step 4
6. 205 n•m [151 ft-lb]
7. Repeat Step 6

Note : The torque only method must only be used when a torque angle dial is inaccessible.

Check the side clearance between the rod and the crankshaft. Move the rods as far as possible toward the crankshaft web.

Measure the side clearance with a feeler gauge.

Connecting Rod Side Clearance		
mm		in
0.30	MIN	0.012
0.51	MAX	0.020



Both rods **must** move freely from side-to-side.